

cwltool

System	No Container	Singularity
LUMI	ERROR (26)	INVALID (250)
Laptop	ERROR (2)	INVALID (202)
Cloud	OK (1135)	ERROR (3)

Table 1: FALL3D workflow using **baseCommand: mpirun**. Execution time is shown in seconds. “INVALID” indicates runs where the MPI job was launched via Singularity instead of the host MPI launcher.

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System	No Container	Singularity
LUMI	OK (219)	ERROR (4)
Laptop	ERROR (2)	ERROR (3)
Cloud	OK (1135)	ERROR (5)

Table 2: FALL3D workflow using **MPIRequirement**. Execution time is shown in seconds.